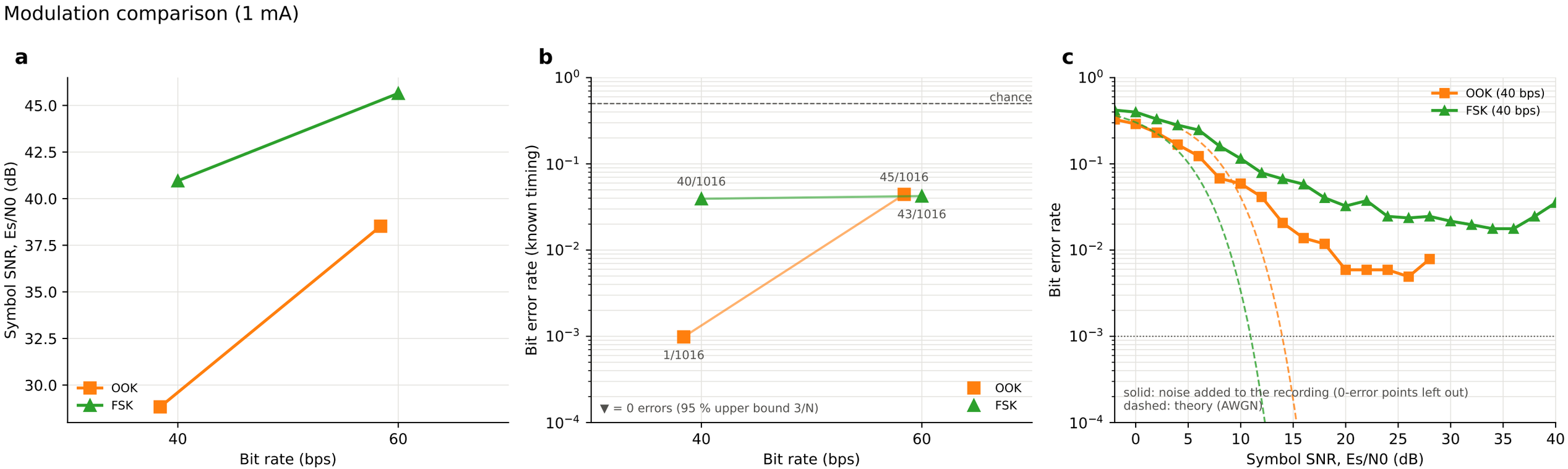
Modulation comparison (1 mA)
a
Symbol SNR, Es/N0 (dB)
45.0
42.5
40.0
37.5
35.0
32.5
30.0
40
60
Bit rate (bps)
OOK
FSK
b
Bit error rate (known timing)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
chance
40/1016
45/1016
43/1016
1/1016
▼ = 0 errors (95 % upper bound 3/N)
OOK
FSK
40
60
Bit rate (bps)
c
Bit error rate
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
OOK (40 bps)
FSK (40 bps)
solid: noise added to the recording (0-error points left out)
dashed: theory (AWGN)
0
5
10
15
20
25
30
35
40
Symbol SNR, Es/N0 (dB)